BUILD MANUAL & ROADMAP LEDGER

# Damn Beavers

A systemic beaver-civilization simulator built on one rule above all others: *systems, not scripted outcomes.* Every mechanic — water, wood, hunger, causality — is a real, inspectable simulation, never a hard-coded story beat.

**Phase 0** ENGINE FOUNDATION **— Passed**

**Phase 1** ONE WATERSHED **— 3 of 22 done**

**Milestone 0** POND TEST **— Not passed**

STATUS AS OF 2026-08-15 · CURRENT TASK P1-04 (Unified Terrestrial/Aquatic Navigation) — IN REVIEW · 406 TESTS PASSING

STA 0+00 — OVERVIEW

| | | | |
|---|---|---|---|
| **15** / 35 TASKS DONE | **406** TESTS PASSING | **34** TEST FILES | **4,457** LINES OF SIM CODE |
| **6,027** LINES OF TEST CODE | **15** / 15 TASKS WITH A CRITIC-FOUND DEFECT | | |

§01

# Where the build stands

Phase 0 laid the engine foundation — clock, identity, components, persistence, debug tooling — and is fully closed out. Phase 1 is building the first watershed: terrain, water, flow, and navigation are done; beavers, needs, and jobs are next.




**Phase 0 — Engine Foundation**

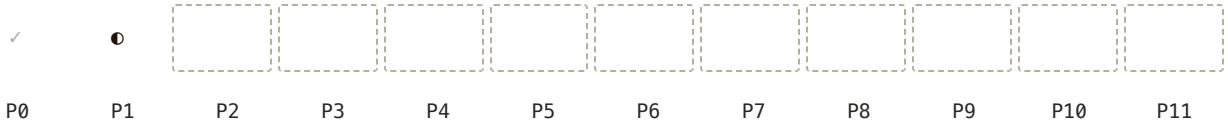
100% · gate PASSED 2026-08-13




**Phase 1 — One Watershed**

14% · P1-04 in review

### Full arc — Engine Foundation through Milestone 0




M0 — Pond Test gate: not reachable until every Phase 1 task is done

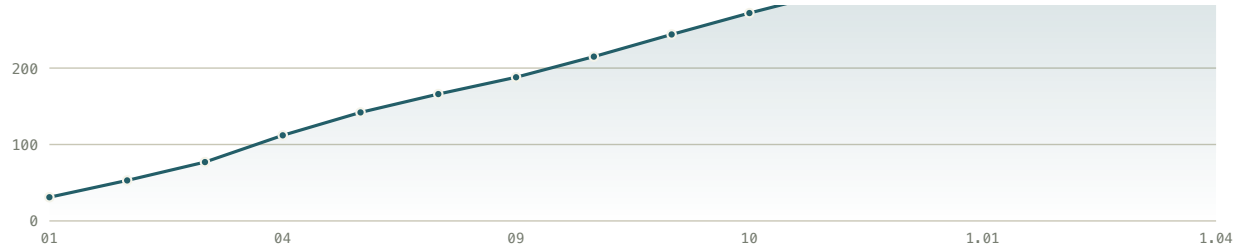
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STA 1+40 — TEST SUITE GROWTH

### Cumulative tests passing, by completion order

Every task adds unit tests and a headless acceptance test before it can close. 31 → 406 across 15 completed tasks; P1-04's tests exist but its critic pass isn't closed out yet.






§02

## The quality process

No task is marked done on the strength of "it compiles." Every feature — no exceptions so far — goes through the same five-step cycle before its ledger row closes.

**01**

### Implement

The smallest correct mechanism for exactly this task — never a future phase's system built early.

**02**

### Unit + acceptance tests

Unit tests for the mechanism, plus a headless test matching the roadmap's literal acceptance wording.

**03**

### Independent critic review

A fresh reviewer with no context tries to break it: correctness, determinism, persistence, generality.

**04**

### Fix + regression-test findings

Substantive findings get fixed, with a new test that fails on the old code and passes on the fix.

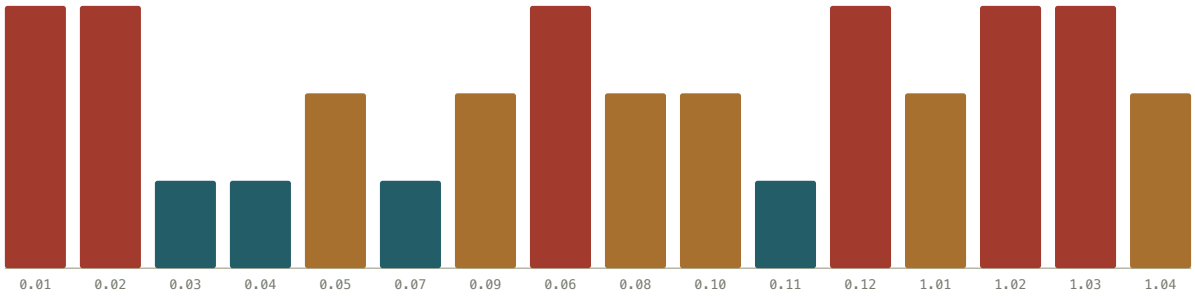
**05**

### Ledger entry with evidence

TASKS.md is updated with test counts and a full account of what the critic found.

### Findings severity, by task

Severity weight of the most serious substantive defect a critic pass found in each completed task (Critical=3, Moderate=2, Minor=1). Two tasks — water volume and channel flow — required a full algorithm replacement.




```
Critical — real correctness bug, wrong results with no error

Moderate — real defect, contained blast radius

Minor — observability, docs, or test-coverage gap
```

**15/15**

Every completed task has had at least one substantive, independently-verified finding fixed before it closed — including two severe bugs (water-volume delivery off by 3.5×, and an undamped flow oscillation) that required rewriting the algorithm, not just patching a check.

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§03

## The ten non-negotiable laws

From `AGENTS.md` — the constitution every task is built and reviewed against.

1.1

### Systems, not scripted outcomes

Never a named story event like `DamFloodEvent`. Implement the underlying cause — water + pressure + flow + obstruction + permeability — and let the outcome emerge.

1.2

### The simulation is authoritative

Renderer, UI, and audio only observe state; they never own it. A mesh is not a tree. The simulation must keep making sense running fully headless.

**1.3**

### Stable persistent identity

Every important entity gets a permanent ID that survives save/load — never render-object identity, never array position.

**1.4**

### Causality must be inspectable

Every important state change can answer: what happened, why, what caused it, what blocked it, who participated, where and when.

**1.5**

### Jobs are intentions, not puppeteering

Designation → work → job → eligibility → agent choice → reservation → navigation → execution → completion. Never teleport an agent or item to satisfy an order.

**1.6**

### Physical resources are physical

Three logs means three log entities that exist, get carried, and get delivered — never an invisible integer counter.

**1.7**

### Hydrology changes navigation through reality

A flooded route becomes impassable because depth, locomotion, slope, and current actually change cost — never `if damBuilt: blockLowRoad()` .

**1.8**

### Determinism where practical

Same version, seed, save, and command stream reproduces the same state. Authoritative nondeterminism is a bug, not a feature.

**1.9**

### Persistence is part of every system

Never bolted on at the end. Every new persistent state considers serialization, versioning, and migration on day one.

**1.10**

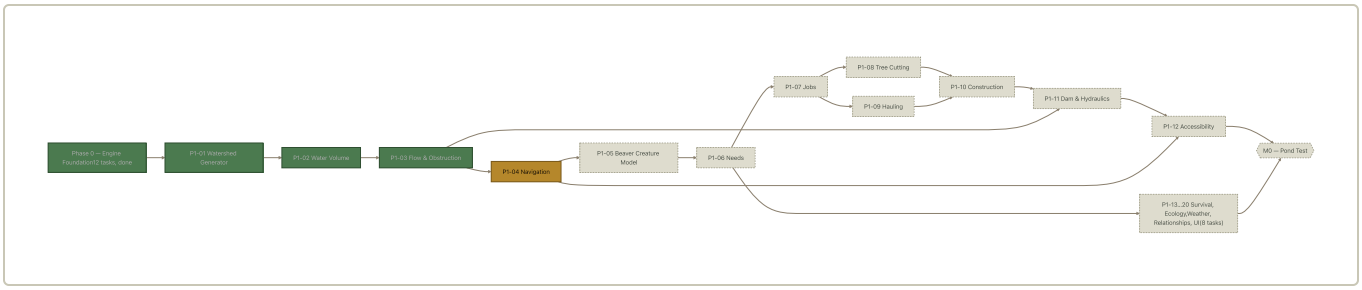
### Do not implement future phases early

Implementation order comes strictly from `TASKS.md` . A later feature may inform an interface — it never gets built ahead of its dependencies.

§04

## Dependency map

The critical path from engine foundation to the Pond Test. A condensed view — the full 35-task dependency graph lives in `TASKS.md`.




§05

## Task ledger

Every task, its status, dependencies, and a condensed account of what shipped and what the critic found. Full evidence paragraphs live in `TASKS.md`.

ALL 35 | DONE · 15 | IN REVIEW · 1 | BLOCKED · 19

### Phase 0 — Engine Foundation 12 / 12 done

| | | | |
|---|---|---|---|
| P0–01 | Simulation clock | DONE | deps: — |
| P0–02 | Stable entity identity | DONE | deps: P0–01 |
| P0–03 | Universal component/state model | DONE | deps: P0–02 |
| P0–04 | Seeded random streams | DONE | deps: P0–01 |
| P0–05 | Spatial world representation | DONE | deps: P0–03 |

| | | | |
|---|---|---|---|
| P0–07 | Material definition system | DONE | deps: P0–03 |
| P0–09 | Terrain and strata | DONE | deps: P0–05, P0–07 |
| P0–06 | Versioned persistence schema | DONE | deps: P0–02, P0–03, P0–05 |
| P0–08 | Event and causality log | DONE | deps: P0–01, P0–02 |
| P0–10 | Plants and trees as physical entities | DONE | deps: P0–02, P0–03, P0–07, P0–09 |
| P0–11 | Headless simulation runner | DONE | deps: P0–01…P0–10 |
| P0–12 | Debug inspection layer | DONE | deps: P0–03, P0–05, P0–08 |

## Phase 1 — One Watershed 3 / 22 done, 1 in review

| | | | |
|---|---|---|---|
| P1–01 | Watershed map generator | DONE | deps: P0–09, P0–10 |
| P1–02 | Water volume | DONE | deps: P0–05, P0–07, P1–01 |
| P1–03 | Flow, drainage, hydraulic obstruction | DONE | deps: P1–02 |
| P1–04 | Unified terrestrial/aquatic navigation | IN REVIEW | deps: P0–05, P1–02, P1–03 |
| P1–05 | Beaver creature model | BLOCKED | deps: P0–03, P1–04 |
| P1–06 | Basic needs | BLOCKED | deps: P1–05, P0–01 |
| P1–07 | Job/order architecture | BLOCKED | deps: P1–05, P1–06 |
| P1–08 | Tree cutting | BLOCKED | deps: P0–10, P1–07 |

| | | | |
|---|---|---|---|
| P1–09 | Physical items and hauling | **BLOCKED** | deps: P0–07, P1–07, P1–08 |
| P1–10 | Construction jobs | **BLOCKED** | deps: P1–07, P1–09 |
| P1–11 | Dam structure and hydraulic interaction | **BLOCKED** | deps: P1–03, P1–10, P0–07 |
| P1–12 | Accessibility recomputation from water | **BLOCKED** | deps: P1–04, P1–11 |
| P1–13 | Basic food and drink | **BLOCKED** | deps: P1–06, P1–07, P1–09 |
| P1–14 | Sleep and shelter preference | **BLOCKED** | deps: P1–06, P1–10 |
| P1–15 | Basic temperature exposure | **BLOCKED** | deps: P0–01, P1–05 |
| P1–16 | Basic injury and death | **BLOCKED** | deps: P1–05, P1–06, P1–15 |
| P1–17 | Minimal ecology response | **BLOCKED** | deps: P0–10, P1–03 |
| P1–18 | Basic weather and day/night | **BLOCKED** | deps: P0–01, P1–01 |
| P1–19 | Relationships baseline | **BLOCKED** | deps: P1–05 |
| P1–20 | Minimal game UI | **BLOCKED** | deps: P0–12, P1–07 |
| P1–21 | Save/reload full watershed state | **BLOCKED** | deps: P0–06, P1–01…P1–20 |
| P1–22 | Causal explanation query | **BLOCKED** | deps: P0–08, P1–11, P1–12, P1–16 |

## Milestone 0 gate, not reachable yet

| | | | |
|---|---|---|---|
| M0 | Pond Test / Milestone 0 gate | **BLOCKED** | deps: P1–01…P1–22 |

§06

# Roadmap ahead

Deliberately not expanded into tasks yet — per Law 1.10, these stay dormant until the Pond Test passes.

**PH.02** **Living Colony**

Beyond seven beavers: population growth, multi-family colonies, and the social scaffolding a real settlement needs.

**PH.03** **Industrial Lodge**

Deeper construction, crafting chains, and infrastructure beyond a single dam and lodge.

**PH.04** **Living Ecosystem**

A fuller ecology — more species, food webs, and environmental feedback loops than M0's minimal proof.

**PH.05** **Society & History**

Relationships, culture, and a persistent historical record of what the colony has done.

**PH.06** **World Simulation**

Beyond one watershed — a wider simulated world the colony exists within.

**PH.07** **War, Politics & Civilization**

Conflict, governance, and civilization-scale systems layered on the systemic foundation.

**PH.08** **Chronicle**

A player-facing history/legends system built on the same causal event log the engine already keeps.

**PH.09** **Wayfarer**

Travel and exploration beyond the home watershed.

**PH.10** **DF Parity Audit**

A deliberate pass checking coverage against Dwarf Fortress–class depth of interaction.

**PH.11** **Systems DF Doesn't Have**

Where Damn Beavers goes past its own inspiration — beaver-specific systems with no DF analogue.

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§07

## Milestone 0 — The Pond Test

Seven beavers arrive beside a stream. The player designates trees and a dam; ordinary AI does the rest. This is the project's first complete playable nucleus — every item below must hold with no scripted shortcuts.

**NOT PASSED** — blocked on Phase 1 tasks P1-05 through P1-22

- [ ] Stream generates and flows
- [ ] Player can designate trees
- [ ] Trees create physical logs
- [ ] Player can order a dam
- [ ] Dam obstructs simulated water
- [ ] Seven persistent beavers arrive
- [ ] Ordinary AI chooses cutting work
- [ ] Ordinary AI hauls logs
- [ ] Physical materials are delivered
- [ ] Upstream depth rises